

Dear Members of the Selection Committee,

I am writing to express my strong interest in the advertised position of Research Associate in “Autonomous Systems and Robotics – Research, Teaching, and Research Management” (Reference 42/26) at the Department of Computer Science of the Bonn-Rhein-Sieg University of Applied Sciences (H-BRS). As an alumnus of the “Autonomous Systems” (MAS) Master’s program, a long-standing researcher at H-BRS, and a project leader, I am highly familiar with both the organizational structures and the participating research groups of the b-it, and maintain an extensive network within the university.

My roots in local robotics research run deep: I successfully completed my research-oriented undergraduate project at the Fraunhofer Institute for Intelligent Analysis and Information Systems (IAIS) in the field of robotics. There, I also gained valuable experience in human-robot interaction within the EU search-and-rescue robotics project NIFTi, where we investigated cooperative teamwork between humans and robots in urban search-and-rescue scenarios. During my subsequent Master’s studies, I was an active member of the b-it-bots team in the RoboCup@Home competition, developing robotic perception systems for domestic robotics. This background has fostered a strong passion for and solid experience in experimental work with physical robot systems at the interface of simulation, optimization, and real-world controllers.

Scientifically, the generation of robust, adaptive behaviors connects my research focus: **Human-AI Interaction**, applied AI, and co-creative decision support. In collaborative projects such as **OpenSKIZZE** (as Principal Investigator/PI), I work on interactive AI for urban planning, while in **CytoTransport** and previously in **UMMBAS**, I develop interactive systems for complex scientific analysis and design tasks. Methodologically, I utilize a broad ML spectrum (including *U-Net*, *GNNs*, *diffusion models*, *Gaussian process regression*, *Random Forests*, and *ensemble learning*) and work extensively with *agentic AI* and LLMs using modern frameworks like *Cline*, *Roo Code*, and *Zoo Code*. The interface between simulation, data-driven optimization (such as Quality-Diversity optimization for generating robust robotic controllers), and physical systems lies at the core of my approach.

One of my key strengths is acting as a bridge builder between the different departments and institutes of H-BRS. I am active in Department 02 (Computer Science), Department 03 (Engineering), and the TREE Institute. Currently, within the CytoTransport project, I work closely with Department 05 (Applied Natural Sciences) and the IFGA Institute. I intend to leverage this interdisciplinary network as Co-Lead of the new “Digital Twin 4 Multiphysics” Laboratory (in cooperation with Prof. Dr. Sebastian Houben and Prof. Dr. Dirk Reith) to forge new connections for collaborative research and applied teaching.

In academic instruction, I bring extensive experience in **English-language, research-oriented teaching**. Since 2016, I have independently taught courses such as “Evolutionary Computation” as well as the interdisciplinary Master’s module “Evolutionary Computation: Theory & Application”

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(ECTA), which specifically bridges the gap between students of Departments 02 and 03. A particular strength of mine lies in the structured supervision, educational development, and assessment of students, reflected in the successful guidance of **over 40 student theses and projects** across diverse domains – frequently in close cooperation with industrial partners like SICK, Mundialis, DrofoTech, Geo-NET, or Kautex Textron.

Regarding the future of robotics education, I firmly believe that applicable domain knowledge is becoming increasingly important. However, without a deep understanding of software implementation, technology risks becoming a black box. The responsibility of a modern university is to “de-blackbox” our students’ minds. Our goal in the MAS program must be to enable students to identify and develop new niches where technological solutions can create broad societal value.

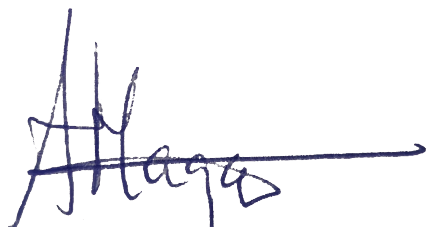
My profile is complemented by solid experience in **research management**. As a project manager of third-party funded projects (such as OpenSKIZZE) and the FullDA-FM project (financed by the start-up grant of the Vice President for Research and Transfer, VP2), I have held budget responsibility, coordinated consortium partners, and supervised student research assistants. With this experience, I aim to support the faculty with administrative tasks, coordinate collaboration with research and industry partners, and actively contribute to drafting new research proposals for external funding.

The opportunity to remain at my academic home, shape the next generation of students, and actively conduct research at the intersection of modern AI and autonomous robotics is exceptionally motivating for me.

Additionally, I would like to mention that Section 1 and Section 7 of the German General Act on Equal Treatment (AGG) apply to me.

Thank you very much for your time and for considering my application. I look forward to the opportunity of a personal interview.

Sincerely,

A handwritten signature in blue ink, appearing to read 'A. Hagg', with a long horizontal stroke extending to the right.

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Dr. Alexander Hagg

Researcher in AI & Optimization

Short Profile

Dutch researcher, born on June 30, 1979, with international experience in research and teaching. My research focuses on the intersection of **Human-AI Interaction**, interactive decision support, and applied AI. In collaborative projects like OpenSKIZZE, CytoTransport, and UMMBAS, I develop interactive systems that support human users in complex planning, analysis, and design tasks. Methodologically, I work with a broad spectrum of applied ML models, such as **U-Net**, **GNNs**, **diffusion models**, **Gaussian process regression**, **Random Forests**, **ensemble learning**, as well as **agentic AI** and LLMs, making intensive use of agent frameworks like **Cline**, **Roo Code**, and **Zoo Code**. Furthermore, I possess solid experience in domestic robotics (b-it-bots), ROS-based robot perception, and search-and-rescue robotics in the context of human-robot interaction (NIFTi project).

Academic Background and Education

2017–2021 **PhD, LIACS, Leiden University**, Leiden, Netherlands

Discovering the Preference Hypervolume: an interactive model for co-creative computation.

Using modern Quality-Diversity optimization methods, Gaussian process models, and generative Variational Autoencoder models, an interactive model was developed enabling collaborative human-AI interaction to explore extremely large solution spaces, identify optimal solutions, and open up the hypervolume containing user-preferred solutions. *Defense: December 7, 2021*

2018 **Gaussian Process Summer School**, *University of Sheffield*, Sheffield, UK

The Gaussian Process Summer Schools are a series of training sessions and workshops for researchers who wish to understand and apply Gaussian process models in theory and practice.

2013–2016 **M.Sc. Autonomous Systems**, *Bonn-Rhein-Sieg University of Applied Sciences*, Sankt Augustin, Germany

Focus on practical methods for autonomous systems, robot perception, and mobile robotics. Active member of the b-it-bots RoboCup@Home team (developing robot perception systems using ROS, C++, Python). Within an independent *R&D project*, developed a perception system capable of identifying transparent household objects based on physical limitations of an RGB-D camera. Master's thesis utilized hierarchical ML methods for dimensionality reduction.

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- 2012 **Summer School Computer Vision**, *Czech Technical University in Prague*, Prague, Czech Republic
Summer school on advanced computer vision and machine learning methods.
- 2011 **Introduction to Artificial Intelligence MOOC**, *Stanford University*, California, USA
MOOC taught by Sebastian Thrun and Peter Norvig. Out of 104,000 registered students, 13,000 successfully completed the course.
- 2009–2012 **B.Sc. Computer Science**, *Bonn-Rhein-Sieg University of Applied Sciences*, Sankt Augustin, Germany
Specialization: Embedded Systems, Computer Vision, and Robotics. Successful Bachelor's project at Fraunhofer IAIS in autonomous mobile robotics; Bachelor's thesis also completed at Fraunhofer IAIS.
- 2006–2009 **Vocational Training**, *City of Troisdorf*, Troisdorf, Germany
IT Specialist for System Integration

Project and Professional Experience

- 2025–present **Co-Lead / Deputy Head**, *Digital Twin 4 Multiphysics Laboratory (A²S)*, Sankt Augustin
Interdepartmental research and teaching cooperation between Prof. Dr. Sebastian Houben (Dept. 02) and Prof. Dr. Dirk Reith (Dept. 03) for the collaborative study of multiphysical digital twins at the interface of simulation and physical robotic systems.
- 2025–present **Research Associate in the CytoTransport Project (DFG)**, *Bonn-Rhein-Sieg University of Applied Sciences*, Sankt Augustin
Interdisciplinary DFG research cluster across Departments 02, 03, and 05, as well as the IFGA and TREE institutes (assigned to Dr. Karl N. Kirschner). My focus is on studying **Human-AI Interaction in scientific applications** for predicting molecular conformations, optimizing force field parameters for simulations, and predicting the impact of mutations in proteins.
- 2024–2025 **Project Leader (Principal Investigator) in the OpenSKIZZE Project**, *Bonn-Rhein-Sieg University of Applied Sciences*, Sankt Augustin
The OpenSKIZZE project in Department 02 (assigned to Prof. Dr. Dirk Reith) serves as a primary example of **interactive, applied AI** and co-creative decision support for generating climate-adapted urban planning designs at the neighborhood level. As Project Leader (PI), I was responsible for implementing interactive, surrogate-assisted multi-solution optimization, integrating the cold air outflow model KLAM_21 from the German Weather Service (DWD), and supervising three students in developing machine learning models to predict cold air currents and heat. For practical application, I developed an interactive prototype application, including workflow and visualization.

Partners: Montag Stiftung Urbane Räume gAg, Neue Stadtgärtnerei e.V.

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2023–2025 **Research Associate in the UMMBAS Project**, *Bonn-Rhein-Sieg University of Applied Sciences*, Sankt Augustin

UMMBAS combined interdisciplinary expertise from Departments 02, 03, and 05, alongside the IFGA and TREE institutes, to develop computational, interactive methods (assigned to Prof. Dr. Dirk Reith). The central focus was on **Human-AI Interaction in scientific applications** and establishing collaborative data analysis in a shared, immersive VR environment. Within this project, I co-supervised work on Lennard-Jones force-field parameterization and generative graph neural networks for molecular conformation generation, and developed the virtual screening framework MolFuSE. I also co-supervised work on visualizing large molecular spaces in VR/AR.

2019–2020 **Project Leader (Principal Investigator) in the FullIDA-FM Project**, *Bonn-Rhein-Sieg University of Applied Sciences*, Sankt Augustin

Research project in Department 02 (in cooperation with Prof. Dr. Dirk Reith), funded by the start-up grant of the Vice President for Research and Transfer (VP2). The project focused on simulation- and optimization-assisted decision support in urban planning to develop an early prototype (including all core machine learning and optimization algorithms) and to secure follow-up funding.

Partners: Montag Stiftung Urbane Räume gAg, Neue Stadtgärtnerei e.V.

2019–2020 **Research Associate in the KISs-BiS Project**, *Bonn-Rhein-Sieg University of Applied Sciences (Dept. 02)*, Sankt Augustin

Research project in Dept. 02 in cooperation with Prof. Dr. Alexander Asteroth. While the benefits of AI and machine learning in sports are undisputed, practical implementation presents significant challenges regarding resource access, expert availability, and method/data handling. A systematic review was authored as a starting point. Based on the review findings, proposals were developed to bridge these gaps, aiming to initiate new transfer and synergy pathways as well as innovative product developments.

Partners: Technical University of Munich, University of Mainz

2016–2019 **Research Associate in the AErOmAt Project**, *Bonn-Rhein-Sieg University of Applied Sciences (Dept. 02)*, Sankt Augustin

Research project in Dept. 02 in cooperation with Prof. Dr. Dirk Reith and Prof. Dr. Alexander Asteroth. The AErOmAt project aimed to develop novel methods to significantly reduce the number of aerodynamic simulations required in computationally intensive optimization domains. Methods for computational creativity, efficient optimization, and problem decomposition were developed, focusing on modern evolutionary, machine-assisted, and numerical methods. These methods were applied to shape optimization for airfoils, recumbent bicycles, a Formula Student racing car, cabin airflow, and passenger car side mirrors.

Partners: The project team was complemented by five associated partners: the German Aerospace Center (DLR), the Fraunhofer Institute for Algorithms and Scientific Computing (SCAI), the University of Siegen, SIDACT GmbH, and Beyss Leichtfahrzeuge (Go-One).

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2014–2015 **Research Associate in the E-LaBoR Project**, *Bonn-Rhein-Sieg University of Applied Sciences (Dept. 02)*, Sankt Augustin

Research project in Dept. 02 in cooperation with Prof. Dr. Stefanie Meilinger, Prof. Dr. Alexander Asteroth, and IZNE. A phased rollout strategy for charging infrastructure in Bonn and the Rhein-Sieg district was developed, taking into account specific regional differences and baseline situations in mobility behavior and potential grid connection points. Additionally, a planning guide was drafted to incorporate further constraints during an exemplary detailed analysis as a post-processing step for selecting individual charging locations previously identified via optimization.

Partners: City of Bonn, Rhein-Sieg District, energy providers SWB Energie und Wasser, rhenag, Stadtwerke Troisdorf, RheinEnergie, and RWE.

2011–2012 **Student Assistant**, *Fraunhofer IAIS*, Sankt Augustin

NIFTi Project. NIFTi researched **search-and-rescue robotics** and **human-robot interaction** in urban search-and-rescue scenarios, particularly during multi-robot deployments to enable cooperative teamwork and reduce the operator's cognitive load. NIFTi operationalized natural collaboration as a trade-off between operational and cooperative demands within a cognitive architecture, aiming to minimize cognitive load and optimize joint team workflow.

Partners: DFKI Language Technology Lab, TNO, Fraunhofer IAIS, Bluebotics, ETH Zurich, Czech Technical University in Prague, Sapienza University of Rome, Dortmund Fire Department, Corpo Nazionale Vigili del Fuoco (Italian Ministry of the Interior)

2002–2006 **Freelancer**, North Rhine-Westphalia

Graphic design, web design, web development.

Skills and Expertise

Methods: Experience in developing, implementing, and interpreting applied Machine Learning ML models, especially with scientific data (e.g., U-Net, GNNs, diffusion models, Gaussian process regression, Random Forests, ensemble learning, as well as agentic AI and LLMs).

Applications Computational Fluid Dynamics (CFD), computational chemistry, interactive urban planning, shape optimization, AI in sports, search-and-rescue robotics (human-robot teamwork), domestic robotics (robot perception), cooperative autonomous systems.

Methods: Expertise in multi-solution Optimization optimization algorithms (e.g., Quality-Diversity optimization; historically developed for robust, adaptive controllers at the interface of simulation, optimization, and physical robotic systems).

Computational Chemistry Experience with force field parameterization, molecular similarity models, pre-screening of small molecules, and conformation optimization.

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Coding	Excellent knowledge of Python (including agentic AI and LLMs using agent frameworks like Cline, Roo Code, Zoo Code) and C/C++ with a focus on scientific methods and experimental robotics. Operating Systems & Middleware: Linux, ROS. Frameworks: PyTorch, scikit-learn, pyribs, pandas, NumPy, SciPy, OpenCV	Data Analysis	Solid knowledge of data analysis, including statistics and visualization, using Python and Jupyter Notebooks.
Research Management	Proven research management competence (Project Leader / PI in OpenSKIZZE and FullDA-FM). Experienced in consortium leadership, budget management, leading scientific teams, and actively contributing to grant applications to support faculty members.	Teaching	Extensive experience in English-language, research-oriented teaching, as well as independent student supervision, assessment, workshop organization, and curriculum development.

Languages

Dutch	Native speaker
German	Fluent
English	Fluent

Grants, Fellowships, and Awards

- 2024 **Grant Approved**, *German Federal Environmental Foundation (DBU)*
Funding approved for the OpenSKIZZE project, supporting an 18-month project
- 2023 **Honourable Mention**, *Best Paper Awards*
GECCO 2023
- 2022 **Honourable Mention**, *ACM SIGEVO Best Dissertation Awards*
ACM SIGEVO
- 2022 **Research Funding**, *Bonn-Rhein-Sieg University of Applied Sciences*
Grant approved for FullDA-FM, funded by the start-up grant of the Vice President for Research and Transfer (VP2) for an 18-month project
- 2021 **Nominated**, *Best Paper Awards*
GECCO 2021
- 2019 **Co-author of Grant Proposal**, *Federal Institute for Sports Science (BISp)*, Sankt Augustin
Approved grant proposal for the KISs-BiS project, duration: 1 year
- 2017–2020 **Full PhD Scholarship**, *H-BRS/NRW*, Sankt Augustin
- 2017 **Award**, *AFCEA*
AFCEA Student Award 2017, 2nd Place

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- 2017 **Honourable Mention, GECCO Conference**
GECCO Student Workshop 2017
- 2016 **Best Paper Award, RoboCup Symposium 2016**
- 2015 **Co-author of Grant Proposal, BMBF, Sankt Augustin**
Approved grant proposal for the AErOmAt project, duration: 3 years
- 2012–2015 **Full Scholarship, German Academic Scholarship Foundation (Studienstiftung des deutschen Volkes), Sankt Augustin**

Teaching Experience

- 2017–2025 **Examiner and Supervisor, H-BRS (Dept. 02 / Dept. 03 / Dept. 05), Sankt Augustin**
Bachelor's and Master's theses as well as R&D projects (independent strength in student supervision, exam administration, and workshop organization)
- 2022–2024 **Lecturer, H-BRS, Sankt Augustin**
Evolutionary Computation (M.Sc. Autonomous Systems, English-taught, research-oriented course)
- 2023 **Workshop Leader, H-BRS, Sankt Augustin**
Workshop on Artificial Intelligence: Large Language Models and Generative AI (didactic design)
- 2023 **Workshop Leader, H-BRS, Sankt Augustin**
International Summer School on Machine Learning
- 2022 **Lecturer, H-BRS, Sankt Augustin**
Low Power Lab for Machine Learning Applications (Dept. 02)
- 2019–2020 **Guest Lectures, IUBH, Düsseldorf**
Artificial Intelligence in Social Work
- 2019–2020 **Guest Lectures, LIACS, Leiden, Netherlands**
Quality-Diversity Algorithms
- 2018 **Workshop Organizer and Lecturer, H-BRS, Sankt Augustin**
Genetic Algorithms for Young Students
- 2017–2019 **Supervision and Examination, H-BRS (Dept. 02 / Dept. 03), Sankt Augustin**
Various B.Sc. and M.Sc. theses
- 2017 **Summer School Organizer and Lecturer, H-BRS, Sankt Augustin**
Summer School on Genetic Algorithms
- 2017 **Lecturer, H-BRS (Dept. 02), Sankt Augustin**
Internal Lecture on Gaussian Process Regression
- 2016 **Lecturer, H-BRS (Dept. 02 / Dept. 03), Sankt Augustin**
Evolutionary Computation: Theory & Application (ECTA) (M.Sc.) – interdepartmental module for computer science and engineering
- 2015–2019 **Lecturer, H-BRS (Dept. 02), Sankt Augustin**
Genetic Algorithms (B.Sc.) & Neuroevolution (B.Sc.)
- 2013–2015 **Student Assistant / Tutor, H-BRS (Dept. 02, MAS), Sankt Augustin**
Autonomous Mobile Robots (M.Sc.)

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- 2012 **Student Assistant / Tutor**, *H-BRS (Dept. 02)*, Sankt Augustin
Algebra and Number Theory (B.Sc.)
- 2011 **Tutor**, *H-BRS (Dept. 02)*, Sankt Augustin
Theoretical Computer Science (B.Sc.)

Supervision of Student Work (Summary)

- Volume Successful primary, secondary, or tertiary supervision of **over 40 student theses and projects** at Bonn-Rhein-Sieg University of Applied Sciences.
- M.Sc. Theses **19 theses** (including the programs *Autonomous Systems* and *Computer Science*).
B.Sc. Theses **13 theses**.
- Project Work **12 research and project works** (including research-focused R&D and Master's projects).
- Collaborations Numerous projects in direct cooperation with well-known industry partners (including Fraunhofer IAIS, Fraunhofer SCAI, SICK, Mundialis GmbH & Co. KG, DrofoTech, Geo-NET, Kautex Textron, inovex, DGUV).

Interests

- Music Playing neoclassical music, improvisation
- Volunteering Initiative Neue Stadtgärtnerei, creating sustainable housing and agriculture in the center of Bonn
- Sports Hiking and cycling
- Philosophy Intelligence, Artificial Intelligence, politics
- Art Co-creative collaboration between artists and AI, generative art, novel writing (aiming for publication in 2027)
- Politics Geopolitics, European & local politics
- Cooking Indonesian cuisine

Other Activities

- Climate Adaptation Working Group, Twins4NRW (2025)
- H-BRS Representative, GeoIT Round Table NRW, GDI NW (2025)
- Speaker at KI-Wrap (AI lectures), H-BRS (2025)
- Organizer of the innovation meeting "Digital Twin for Urban Sustainability", H-BRS (2025)
- AI Consultant in an Academic Search Committee, H-BRS (2025)
- Reviewer for the Journal of Evolutionary Computation, MIT Press (2024–present)
- Neue Stadtgärtnerei Bonn, a grass-roots, socially and ecologically sustainable housing, agricultural, and educational project. Responsible for urban planning, architecture, and cooperation with the City of Bonn (2022–present)
- Joint Presentation, MPIC and International Center for Sustainable Development (IZNE) on climate adaptation (2023)

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- Guest on the H-BRS research podcast “Abenteuer Promotion” on AI research (2021)
- Program Committee of the PPSN Conference (2020–2023)
- Reviewer for the International Journal of Human-Computer Interaction (2019)
- Reviewer for the European Conference on Machine Learning and Principles and Practice of Knowledge Discovery in Databases (2018)
- Reviewer for IEEE Transactions on Cybernetics (2018–present)
- Panel Discussion “Fair Mobility Bonn” (2016)
- Network development and fundraising for Jugend Rettet / luventa (2016)
- Participant in RoboCup@Home selections with the “b-it-bots” team; developing perception systems for domestic robotics using ROS, C++, and Python (2012–2015)
- Participant in RoboCup German Open, Magdeburg (2013)
- Participant in Robolympics, Moscow, Russia (2012)
- Participant in EuRobotics Challenge, London, United Kingdom (2011)

Publications

- Hagg, A., Reith, D., Preller, M., Kirschner, K. N. (2026). *Molecular Function-guided Unsupervised Similarity-based Enrichment for Low-Data Virtual Screening*. Journal of Cheminformatics (under review) 2026.
- Hagg, A., Guerrero, T., Reith, D. (2026). *U-Net-Accelerated Quality-Diversity Optimization for Climate-Adaptive Urban Layouts*. Parallel Problem Solving using Nature 2026.
- Strickstock, R., Hagg, A., Reith, D., Kirschner, K. N. (2025). *Speed up Multi-Scale Force-Field Parameter Optimization by Substituting Molecular Dynamics Calculations with a Machine Learning Surrogate Model*. Journal ChemPhysChem.
- Hagg, A., Gaier, A., Wilde, D., Asteroth, A., & Reith, D. (2025). *Full Domain Analysis in Fluid Dynamics*. Journal of Machine Learning and Knowledge Extraction.
- Strickstock, R., Hagg, A., Hülsmann, M., Kirschner, K. N., & Reith, D. (2025). *Fine-Tuning Property Domain Weighting Factors and the Objective Function in Force Field Parameter Optimization*. Journal of Molecular Graphics and Modelling.
- Spelten, P., Reiswich, A., Kleinert, J., & Hagg, A. (2024). *Demonstrating Sensitivity Analysis using Parametric Urban Design and a GPU-based Lattice-Boltzmann Solver*. ECCOMAS 2024, Lisbon.
- Hagg, A., & Kirschner, K. N. (2023). *Open-Source Machine Learning in Computational Chemistry*. Journal of Chemical Information and Modeling.
- Scarton, L., & Hagg, A. (2023). *On the Suitability of Representations for Quality Diversity Optimization of Shapes*. arXiv preprint arXiv:2304.03520.
- Hagg, A., Kliemank, M. L., Asteroth, A., Wilde, D., Bedrunka, M. C., Foysi, H., & Reith, D. (2023). *Efficient Quality Diversity Optimization of 3D Buildings through 2D Pre-optimization*. Evolutionary Computation, 1-21.
- Müller, M., Hagg, A., Strickstock, R., Hülsmann, M., Asteroth, A., Kirschner, K. N., & Reith, D. (2023). *Determining Lennard-Jones Parameters Using Multiscale Target Data through Presampling-Enhanced, Surrogate-Assisted Global Optimization*. Journal of Chemical Information and Modeling.
- Hammes, F., Hagg, A., Asteroth, A., Link, D.: *Artificial Intelligence in Elite Sports – A Narrative Review of Success Stories and Challenges*. Frontiers in Sports and Active Living, 4, 2022.
- Hagg, A., Asteroth, A., Rasche, C., Bach, K., & Pfeiffer, M. (2021). *Künstliche Intelligenz für den Spitzensport im Spannungsfeld zwischen Big und Small Data:(KISs-BiS)*. Sportverlag Strauß.
- Hagg, A., *Discovering the preference hypervolume: an interactive model for real world computational co-creativity* (dissertation), Leiden, 2021.
- Hagg, A., Berns, S., Asteroth, A., Colton, S., Bäck, T.: *Expressivity of parameterized and data-driven representations in quality diversity search*. The Genetic and Evolutionary Computation Conference GECCO 2021.
- Hagg, A., Asteroth, A., Pfeiffer, M., Hammes, F., Link, D.: *Einsatzmöglichkeiten und Transfer von Künstlicher Intelligenz im internationalen Spitzensport, zwischen Small und Big Data*. Spinfortec 2020 digital, 26, 2020.

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Hammes, F., Link, D., Lames, M., Hagg, A., Asteroth, A., Pfeiffer, M.: *Einsatz von Künstlicher Intelligenz im internationalen Spitzensport. Eine Erhebung des Status Quo*. Spinfotec 2020 digital, 28, 2020.
 Hagg, A.: *Phenotypic Niching using Quality Diversity Algorithms*. Chapter in "Multimodal Optimization by Means of Evolutionary Algorithms, part II" 2020.
 Hagg, A., Preuss, M., Asteroth, A., Bäck, T.: *An Analysis of Phenotypic Diversity in Multi-Solution Optimization*. Bioinspired Optimization Methods and their Applications BIOMA 2020
 Hagg, A., Asteroth, A., Bäck, T.: *A Deep Dive Into Exploring the Preference Hypervolume*. International Conference on Computational Creativity ICC3 2020
 Hagg, A., Wilde, D., Asteroth, A., Bäck, T.: *Designing Air Flow with Surrogate-assisted Phenotypic Niching*. Parallel Problem Solving using Nature 2020.
 Asteroth, A., Hagg, A., Meng, J., Priesnitz, A., Prochnau, L., Reith, D.: *AErOmAt Final Report*. 2020.
 Hagg, A., Asteroth, A., Bäck, T.: *Modeling User Selection in Quality Diversity*. The Genetic and Evolutionary Computation Conference GECCO 2019.
 Hagg, A., Zaefferer, M., Stork, J., Gaier, A.: *Prediction of neural network performance by phenotypic modeling*. The Genetic and Evolutionary Computation Conference GECCO 2019.
 Hagg, A., Asteroth, A., Bäck, T.: *Prototype Discovery using Quality-Diversity*. Parallel Problem Solving using Nature PPSN 2018.
 Hagg, A.: *Hierarchical Surrogate Modeling for Illumination Algorithms*. The Genetic and Evolutionary Computation Conference GECCO 2017.
 Hagg, A., Mensing M., Asteroth A.: *Evolving Parsimonious Networks by Mixing Activation Functions*. The Genetic and Evolutionary Computation Conference GECCO 2017.
 Spieker H., Hagg, A., Gaier, A., Meilinger, S., Asteroth, A.: *Multi-stage evolution of single-and multi-objective MCLP*. Soft Computing 2016.
 Hagg, A., Hegger, F., Plöger, P.: *On Recognizing Transparent Objects in Domestic Environments Using Fusion of Multiple Sensor Modalities*. RoboCup International Symposium 2016.
 Asteroth, A., Hagg, A.: *How to successfully apply genetic algorithms in practice: Representation and parametrization*. International Conference on INnovations in Intelligent SysTems and Applications INISTA 2015.
 Hagg, A., Spieker, H., Oslislo, A., Jacobs, V., Asteroth, A., Meilinger, S.: *Methodische Grundlegung für eine Strategie zum sukzessiven Ausbau der Ladeinfrastruktur für Elektromobilität in Bonn und dem Rhein-Sieg-Kreis*, 2015.
 Spieker, H., Hagg, A., Asteroth, A., Meilinger, S., Jacobs, V., Oslislo, A.: *Successive evolution of charging station placement*. International Conference on INnovations in Intelligent SysTems and Applications INISTA 2015.

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Appendix: List of Supervised Student Works

Ongoing Theses and Projects (Selection)

- 2026 **Master's Thesis**, *Tania Guerrero*
Topic: Precise prediction of urban cold airflow around buildings (Project: OpenSKIZZE)
- 2026 **Bachelor's Thesis**, *Lorin Kanoglu*
Topic: Multi-objective optimization of force field parameters for adapting energy and density in molecular simulations (Project: CytoTransport)
- 2026 **Bachelor's Thesis**, *Alessandro Castielli*
Topic: Conceptualization of the automated optimization of Lennard-Jones parameters using Nondominated Sorting Genetic Algorithm II (Project: CytoTransport)
- 2025 **Master's Thesis**, *Hari Krishna*
Topic: Variant Effect Predictor (VEP) - ML model to predict variant effects on nicotinic receptors (Project: CytoTransport)
- 2025 **Master's Thesis**, *Ragini Mishra*
Topic: ML model to predict heat hotspots in urban areas (Project: OpenSKIZZE)
- 2025 **R&D Project**, *Saloni Pathak*
Topic: Force-field Parameter Optimization using Quality-Diversity Algorithms (Project: CytoTransport)

Completed Theses and Projects (Selection)

- 2025 **Master's Thesis**, *Shashwat Pandey*
Topic: On-Chip Multi-Modal Perception for Industrial Safety using 3D Time-of-Flight Camera on Hailo-8 NPU (Cooperation with SICK)
- 2025 **R&D Project**, *Tania Guerrero*
Topic: Efficiently predicting air flow of massing designs in urban planning using deep learning (Project: OpenSKIZZE)
- 2025 **Master's Thesis**, *Noah Plingen*
Topic: Impact of Data and Feature Scope on Variant Effect Predictors for ENaCs (Project: CytoTransport)
- 2024 **Master's Thesis**, *Maximilian Mensing*
Topic: Conditional GenAI model to generate optimal urban planning designs based on QD data (Project: OpenSKIZZE)
- 2024 **Master's Project**, *Marc Prismak*
Topic: DGUV Fahrradschwingungen - whole-body vibrations on bicycles for delivery riders (Cooperation with DGUV)
- 2023 **Project Work**, *Tobias Streckel*
Topic: PV2Rad calibration - solar radiation sensor calibration
- 2021 **Master's Thesis**, *Vishwas Sharma*
Topic: Learning simulation for plastic sheet extrusion in blow molding machine (Cooperation with Kautex Textron)
- 2019 **Master's Thesis**, *Khaled Alkady*
Topic: Autonomous Aerobatic Maneuvering Under Crosswind Conditions - Digital Fireworks/Drone Light Shows
- 2019 **Master's Thesis**, *Maximilian Müller*
Topic: Efficient Global Multiscale Optimization of Lennard-Jones Parameters for molecular simulations
- 2019 **Bachelor's Thesis**, *Niklas Köhn*
Topic: VE data analysis (velomobile measurements)
- 2019 **Bachelor's Thesis**, *Oleg Bayer*
Topic: Route optimization for couriers in the Arab region - TSP/VRP using genetic algorithms (Cooperation with Aramex)

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- 2018 **Master's Thesis**, *Marc Müller*
Topic: System reliability analysis - Markov chains, FTA, Neural Networks for hazard cataloging
- 2018 **Bachelor's Thesis**, *Sascha Obst*
Topic: Optimization of a velomobile wheel arch cover using surrogate-assisted illumination
- 2018 **Bachelor's Thesis**, *Sergej Kritschmann*
Topic: Speciation in neuroevolution
- 2017 **Master's Thesis**, *Devvrat Arya*
Topic: Algorithm development for radiation maps and optimization
- 2017 **Master's Thesis**, *Jan Kruska*
Topic: Incorporation of constraints in divergent optimization methods - surrogate-assisted illumination
- 2017 **Bachelor's Thesis**, *Jan Reinartz*
Topic: Comparison of univariate and multivariate time series forecasting models using logistics as an example
- 2017 **Bachelor's & Master's Thesis**, *Jonas Meier*
Topic BA: Game Design with Biofeedback and SAIL; Topic MA: Automated Feature Engineering (Cooperation with inovex GmbH)
- 2017 **R&D Project**, *Wei-Chan Hsu*
Topic: SAIL Robot Recovery: Data-Efficient Behavioral Repertoire Learning for Damage Recovery
- 2016 **Master's Thesis**, *Ahmad Drak*
Topic: DoveCopter: A tracking and following platform for the velomobile using hexacopter
- 2016 **Project Work**, *David Uhlig*
Topic: Optimizing Classroom Schedules with Single Objective Genetic Algorithm
- 2016 **Master's Thesis**, *Helge Spieker*
Topic: Combination of Evolutionary Algorithms and Dynamic Programming for Continuous Control Problems
- 2016 **R&D Project**, *Kamran Mahmood*
Topic: Autonomous following of ground vehicle by Hexcopter for velomobile tracking
- 2016 **Seminar Paper**, *Kevin Kapitzka*
Topic: Modelling adaption to physical training - Fitness-Fatigue model
- 2016 **R&D Project**, *Petr Lukin*
Topic: CFD simulations for velomobiles using OpenFOAM; quadcopter control
- 2016 **Master's Thesis**, *Sogol*
Topic: Charging Station Placement for e-mobility; SAIL algorithm development
- 2016 **Bachelor's Thesis**, *Stefan Ziegenhohn*
Topic: Optimization of a velomobile suspension using genetic algorithms
- 2015 **R&D Project**, *Arka Malik*
Topic: Machine learning classification (SVM, Decision Trees, FFNN) for laminar vs turbulent flow detection
- 2015 **Master's Thesis**, *David Schäfer*
Topic: Training Planning Portal for Evaluation of Training Models - automatic training plans for cyclists
- 2015 **Master's Thesis**, *Ludovico Scarton*
Topic: Surrogate Modeling Techniques for Quality Diversity Optimization - voxel-based urban planning with OpenFOAM
- 2015 **Master's Thesis**, *Max Schachtschabel*
Topic: Velo-Messsystem (Velomobile measurement system)
- 2015 **Master's Thesis**, *Peter Flock*
Topic: VeloScan - 3D reconstruction of velomobile surfaces using low-cost sensors and robot arm

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- 2015 **Bachelor's Thesis**, *Thomas Jonas*
Topic: Quality Diversity algorithms for generating training profiles for cyclists
- 2015 **Bachelor's Thesis**, *Tobias Krieger*
Topic: Exploratory Analysis of Evolutionary Algorithms for Force Field Parameter Optimization
- 2015 **Bachelor's Thesis**, *Zied Sassi*
Topic: Identification and optimization of commissions in B2B sales (Cooperation with Ranger)
- 2012 **Master's Thesis**, *Michael Wahlen*
Topic: TuftTracer - automated flow visualization analysis for velomobiles using computer vision
- 2009 **Bachelor's Thesis**, *Maximilian Mensing*
Topic: Evolving Aerodynamic Surrogate Models for 2D PARSEC Airfoils using Neuroevolution
- 2009 **Bachelor's Thesis**, *Dirk Rusche*
Topic: Combination of neuroevolution methods with gradient descent methods for feed-forward networks (Awarded)

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